H.K.E SOCIETY'S

POOJYA DODDAPPA APPA COLLEGE OF ENGINEERING, KALABURAGI

CIE-2 IMPROVEMENT TEST FOR B. E FIRST SEMESTER

Sub Code: CSPHY102

Sub: QUANTUM PHYSICS & APPLICATIONS *DATE: 1.1.2026*

MAX. MARKS: 40 DURATIONS: 75 MIN

NOTE: **Answer the following questions:**

Given physical constant: h = 6.63 x 10^{-34}Js, c = 3 x 10^{8}ms, k = 1.38 x 10^{-23}J/K, mass of electron = 9.1 x 10^{-31}Kg

1.	a. Discuss the failures of classical free electron theory.	CO3	8M
	b. Discuss the variation of fermi factor with respect to temperature and energy with neat diagram.	CO3	8M
	c. An n- type Germanium sample has a Donor density of $10^{21}m^{-3}$. It is arranged in a Hall experiment having magnetic field of 0.5 T and the current density is 500 A/m^2. Find the Hall voltage if the sample is 3 mm wide.	CO3	8M
2.	a. Explain Critical Field and Critical current. Explain Meissner effect with neat diagram?	CO4	8M
	b. Using M-H graphs explain in detail Soft and Hard Superconductors.	CO4	8M
	c. The transition temperature for lead is 7.26K. The maximum critical field for the material is 8x10^{5}A/m. Lead has to be used as superconductor to a magnetic field of 3x10^{4}A/m. What precaution must be taken?	CO4	4M

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B.E. First Semester Examination – January/February - 2026

Quantum Physics and Applications

(Common for CSE, CSD, ISE and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Given Physical constants: Electron charge, $e = 1.6 \times 10^{-19}$ C; Velocity of light, $c = 3 \times 10^{8}$ ms^{-1}; Plank's constant, $h = 6.63 \times 10^{-34}$Js; Boltzmann constant, $K = 1.38 \times 10^{-23}$ J/K; Electron mass, $m_e = 9.11 \times 10^{-31}$ kg; Mass of neutron $= 1.67 \times 10^{-27}$ kg; Permittivity of vaccum, $\varepsilon_0 = 8.85 \times 10^{-12}$Fm^{-1}; Avogadro number, $N_A = 6.025 \times 10^{26}$/Kmole;

Module – I

1. a) Illustrate the construction and working of GaAs laser with a neat sketch and energy level diagram. (08 Marks)
   b) Derive the expression for the numerical NA and θ_0 of an optical fibre, with the help of a neat labeled diagram. (08 Marks)
   c) The ratio of population of two energy levels is 1.059×10^{-30}. What is the wavelength emitted at 330 K? (04 Marks)

2. a) Obtain the expression for energy density of radiation using Einstein's co-efficient A and B thus conclude $B_{12} = B_{21}$. (08 Marks)
   b) Describe attenuation in an optical fiber and explain the various fiber losses. (08 Marks)
   c) Calculate the V-number and number of modes supported by an optical fiber of core radius 25 μm, operating at wavelength 1.3 μm with NA = 0.2. (04 Marks)

Module – II

3. a) Set up one dimensional time independent Schrodinger wave equation. (08 Marks)
   b) Show that the electron emitted during β decay does not pre-exist inside the nucleus using uncertainty principle. (08 Marks)
   c) Calculate first two energy eigen values in an one-dimensional infinite well of width 1 nm. (04 Marks)

4. a) Using the time-independent Schrodinger wave equation for the particle in an infinite well to arrive at an expression for the possible energy eigen values. (10 Marks)
   b) State Heisenberg's uncertainty principle and mention its physical significance. (06 Marks)
   c) Calculate the energy required to rise from ground state to 4th excited state for an electron in one dimensional potential well of width 0.1 nm. (04 Marks)

Module – III

5. a) Explain the failures of classical free electron theory. (08 Marks)
   b) Define Fermi-factor. Discuss the variation of Fermi factor with respect to temperature and energy with neat diagram. (08 Marks)
   c) The concentration of electron at room temperature in an extrinsic semiconductor used in Hall experiment is 1.7×10^{29}m^{-3}, If a current of 100 A flows through the semiconductor strip of thickness 1 mm and the strength of the magnetic field perpendicular to the strip is 3.6 wb/m^2, estimate the hall co-efficient and hall voltage. (04 Marks)

6. a) Explain how QFET overcome the failure of CFET. (08 Marks)
   b) Derive the expression for electron concentration in intrinsic semiconductor. (08 Marks)
   c) An n-type Germanium sample has a Donor density of 10^{21}m^{-3}. It is arranged in a Hall experiment having magnetic field of 0.5 T and the current density is 500 A/m^2. Find the Hall voltage if the sample is 3 mm wide. (04 Marks)

Module – IV

7. a) Explain Meissner effect with neat diagram. (06 M
   b) Write the application of super conductors. (04 M
   c) Calculate the wavelength of laser beam, if the distance between grating element and screen is 150 cm and the first order and second order diffraction maxima is 10 cm and 20 cm.(Given diffraction grating having 2500 lines per inch (LPI)) (10 M

8. a) Using M-H graphs explain in detail soft and hard superconductors. (08 M
   b) Explain DC and AC Josephson effect. (06 M
   c) Calculate Planck's constant of LED, if the knee voltage of yellow and blue 1.71 V and 2.40 V (Given wave length of yellow and blue is 585 nm and 430 nm). (06 M

Module – V

9. a) Explain character animation of jumping with its parts. (06
   b) Explain the strides and strips in ability to walk. (04
   c) Calculate the Fermi energy of copper wire for the given data below with constant current of 63 mA. (Constant 5×10^{-13})

Temperature in °C	Voltage in mV
30	20.9
35	21.2
40	21.5
45	21.9
50	22.2
55	22.5
60	22.9

(10

10. a) Elucidate the importance of size and scale and weight and strength in animation. (05
    b) Illustrate the odd rule and odd rule multipliers with a suitable example. (05
    c) Calculate the magnetic field induction by tangent law and experimentally on the axis of coil for a current flowing across the coil is 0.5 A and 1A, if the mean deflection in magneto deflectometer is 22 and 42 degrees. [Given number of turns = 5, radius of coil = 9.8 cm, $B_e = 0.3810^{-4}$T, $\mu_0 = 4\pi 10^{-7}$T]. (10

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B.E. First Semester Examination – January 2025

Physics for CSE Stream

(Common to CSE, CSD, ISE and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Physical constants:
Velocity of light, C = $3x10^8 ms^{-1}$: Planck's constant, Boltzmann constant, K = $1.38x\ 10^{-23} J/K$:
Avagadaro number $N_A = 6.023 \times 10^{26} qm/kmole$;
Permittivity of Vacuum, $\varepsilon_o = 8.85x10^{-12} Fm^{-1}$: Mass of Neutron, = $1.67\ x10^{-27} kg$,
Acceleration due to gravity g = $9.8\ ms^{-2}$.

Module – I

1. a) Illustrate the construction and working of (GaAs) semiconductor diode laser with suitable diagram. **(09 Marks)**
   b) Derive the expression for numerical aperture and arrive the condition for propagation of light in an optical fibre. **(08 Marks)**
   c) Find the ratio of population of two energy level in a medium in thermal equilibrium, if the wavelength of light emitted at 330K is 632.8 nm. **(03 Marks)**

2. a) With neat diagram explain the different types of optical fibers. **(09 Marks)**
   b) Explain bar code scanner with neat diagram. **(07 Marks)**
   c) The attenuation of light in an optical fiber is estimated at 2.2 dB/km. What fractional initial intensity remains after 2 km and 6 km? **(04 Marks)**

Module – II

3. a) Develop on dimensional time-independent schrodinger wave equation. **(09 Marks)**
   b) Explain Heisenberg's uncertainity principle. Give its physical significance. **(07 Marks)**
   c) Calculate the zero point energy for an electron in a box of width 10 Å. **(04 Marks)**

4. a) Illustrate the non-existence of electrons in the nucleus on the basis of beta-decay observation using Heisenberg uncertainty principle. **(08 Marks)**
   b) Derive an expression for de-Broglie wave length using the concept of group velocity. **(08 Marks)**
   c) A particle of mass 0.65 MeV/c^2 has free energy 120 eV. Find its de Broglie wavelength. C is the velocity of light. **(04 Marks)**

Module – III

5. a) Explain in detail polarization mechanisms in dielectrics. **(10 Marks)**
   b) Explain the frequency dependence of polarization with neat sketch. **(06 Marks)**
   c) A parallel plate capacitor of area 650 mm^2 and a plate separation of 4 mm has a charge of $2 \times 10^{-10} C$ on it. What is the resultant voltage across the capacitor when a materials of dielectric constant 3.5 is introduced between the plates. **(04 Marks)**

6. a) Derive an expression for internal field in case at one dimensional array of atoms in dielectric solids. **(09 Marks)**
   b) Derive Clausius-Mossotti equation. **(08 Marks)**
   c) The electronic polarization of Krypton gas is $3.54 \times 10^{-40} Fm^2$. If gas contain $2.7 \times 10^{25}/m^3$ at NTP, calculate its dielectric constant. **(03 Marks)**

Module – IV

7. a) Obtain an expression for density of states for energy level in a band. **(10 Marks)**
   b) Describe the experimental procedure to find the Planck's constant using an LED. **(10 Marks)**

8. a) Explain type I and type II superconductors. (08 Marks)
   b) The Fermi level in silver is 5.5 eV. What are the energies for which the probabilities of occupancy at 300 K is 0.99. (Given $1J = 6.24 \times 10^{18}eV$). (05 Marks)
   c) Calculate the wavelength of laser beam, if the distance between the grating element and screen is 150 cm and the first order and second order diffraction maxima is 10 cm and 20 cm (Given N = 2500 lines per inch (LPI)) (07 Marks)

Module – V

9. a) Discuss the sailent features of normal distribution using Bell curves. (06 Marks)
   b) Explain the strides and steps in ability to walk. (04 Marks)
   c) Determine the dielectric constant of capacitors using RC circuit. If the length and breadth of capacitor is 47 mm and 5 mm and separation between them is 0.075mm given $R = 100\ k\Omega$. (10 Marks)

Time in seconds	Charging	Discharging
0	0	10.3
10	6.1	4
20	8.6	1.6
30	9.6	0.6
40	10	0.2
50	10.2	0.1
60	10.3	0.0

10. a) Explain description and inferential statistics. (05 Marks)
    b) Explain character animation of jumping with its parts of jump. (05 Marks)
    c) Calculate the Fermi energy of copper wire for a given data below with constant 5×10^{-13}. (10 Marks)

Temperature in K	Voltage mV	Current mA
303	20.9	47
308	21.2	47
313	21.5	47
318	21.9	46.9
323	22.2	46.9
328	22.5	46.9